LFT MORPHISMS WITH QCQS SOURCE ARE LIMITS OF FINITE PRESENTATION MORPHISMS

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Editorial note. No omissions, hints or 'immediate' steps in the counted span.

Allowed standard prerequisites (editorial).

- Finite type closed in fp. Limits, Lemma lemma-finite-type-closed-in-finite-presentation (same chapter)
- Closed is limit closed and fp. Limits, Lemma lemma-closed-is-limit-closed-and-finite-presentation (same chapter)

Source reference. Original lemma 01ZG: sources/stacks-chow/tags.txt, source lines 2591–2591. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01KU: sources/stacks-chow/tags.txt, source lines 2063–2063. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 09ZP: sources/stacks-chow/tags.txt, source lines 12607–12607. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01TR: sources/stacks-chow/tags.txt, source lines 2384–2384. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Assume*

(1) *The morphism f is of locally of finite type.*
(2) *The scheme X is quasi-compact and quasi-separated, and*
(3) *The scheme S is quasi-separated.*

Then $X = \lim X_i$ where the $X_i \to S$ are of finite presentation, the X_i are quasi-compact and quasi-separated, and the transition morphisms $X_{i'} \to X_i$ are closed immersions (which implies that $X \to X_i$ are closed immersions for all i).

Proof. By Lemma 01ZG there is a closed immersion $X \to Y$ with $Y \to S$ of finite presentation. Then Y is quasi-separated by Schemes, Lemma 01KU. Since X is quasi-compact, we may assume Y is quasi-compact by replacing Y with a quasi-compact open containing X. We see that $X = \lim X_i$ with $X_i \to Y$ a closed immersion of finite presentation by Lemma 09ZP. The morphisms $X_i \to S$ are of finite presentation by Morphisms, Lemma 01TR. □

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